

Lecture 7: Scattering from rough surfaces

Part 1: Radar equation, surface characterization, and scattering from undulating surfaces

Physical principles of remote sensing
(Ge 158)
Fall 2026
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Remote sensing image of the day

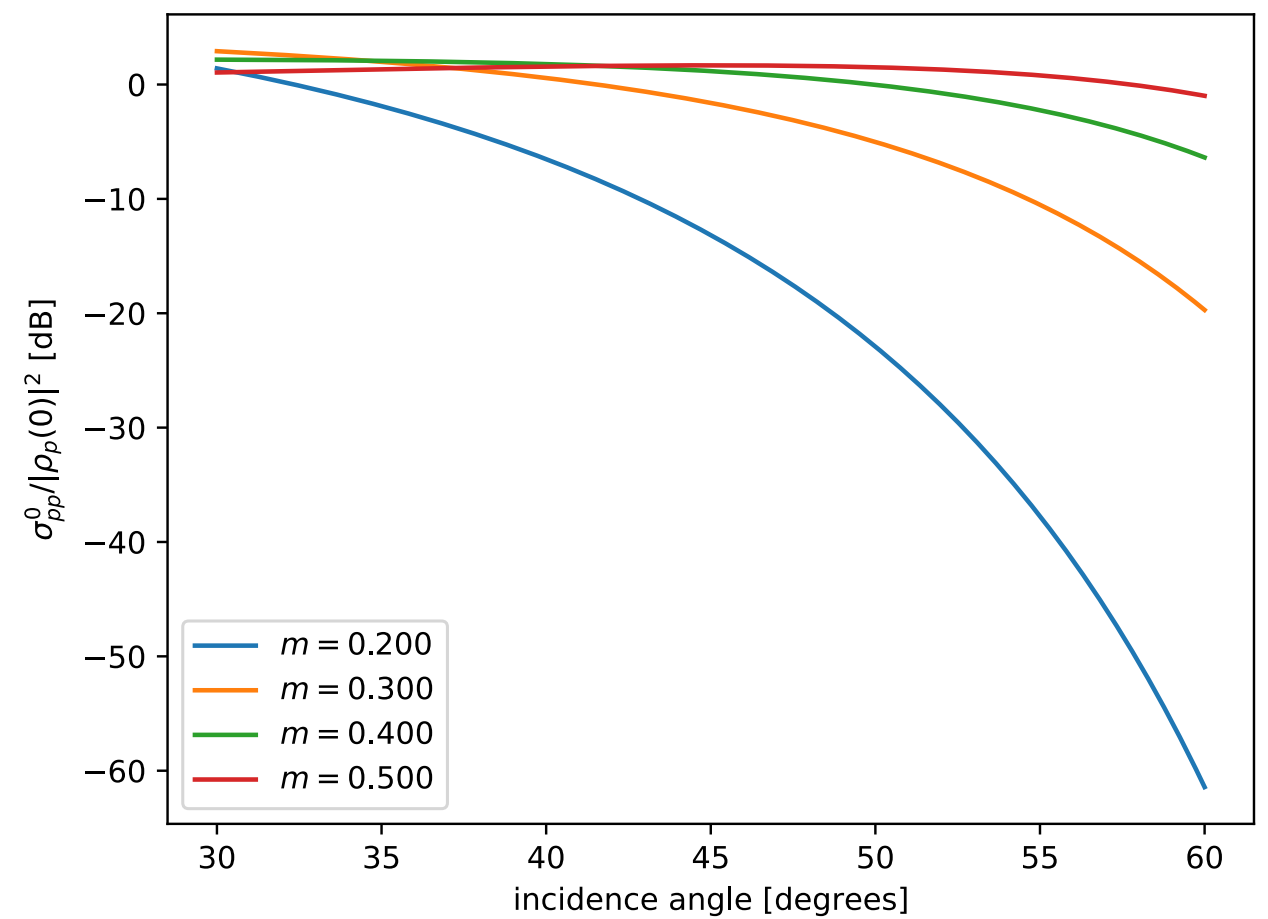
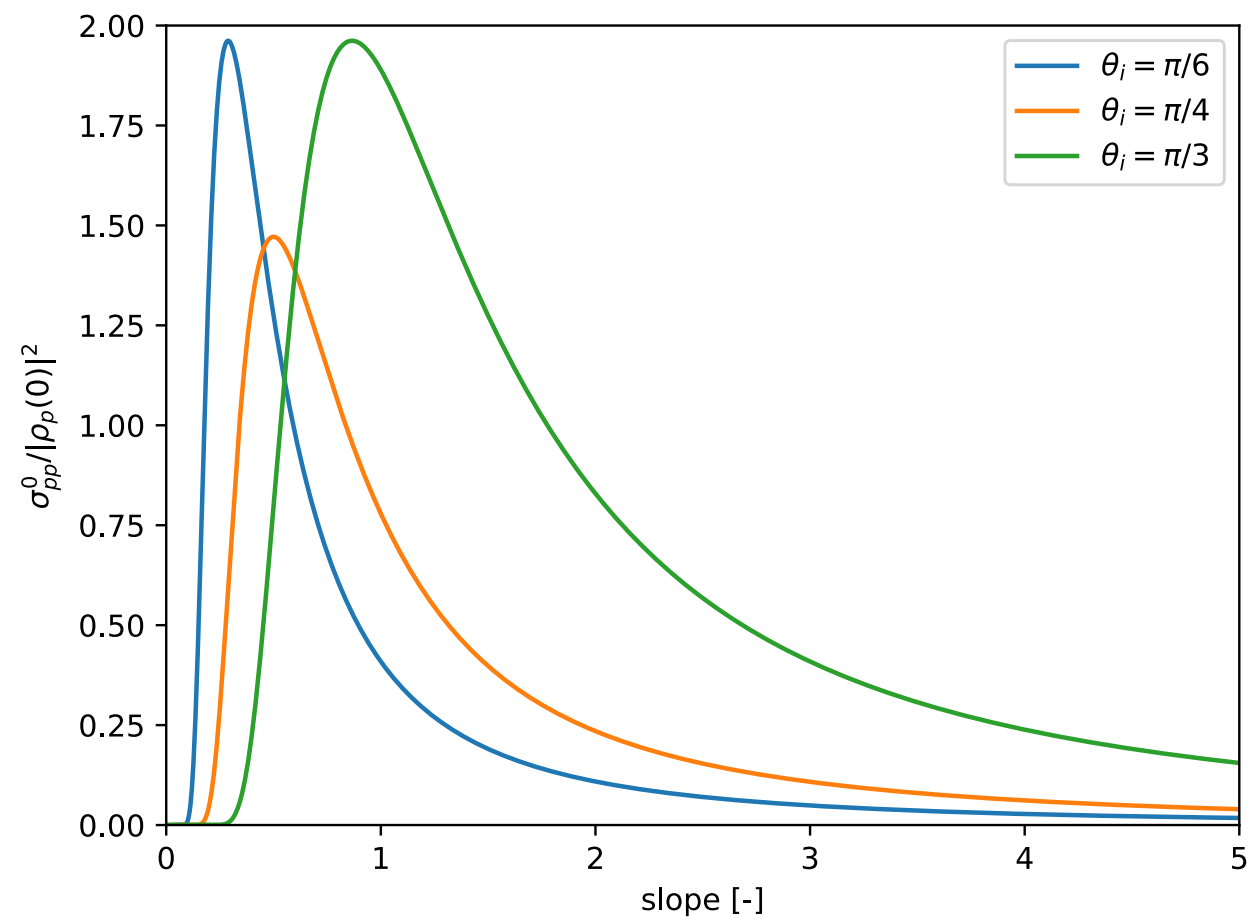


Polarimetric SAR image of Mount St. Helens taken by the UAVSAR instrument. Colors indicate backscatter amplitude at HH (red), HV (green), and VV (blue) polarizations.

Topics covered (see handwritten notes on Stellar)

- Derive radar equation:
 - reinforces material from previous lecture
 - useful for SAR observations
 - sets up discussions of volume scattering and radiometry
- Approximate models for scattering from rough surfaces when roughness is much smaller or much larger than EM wavelength
- Discuss geometric properties of surfaces
- Scattering from an undulating surface

Scattering from undulating surfaces



note that the y-axis is on a linear scale in the lefthand plot and log scale on the righthand plot

Radar scattering from the ocean

backscattered power vs incidence angle is very different from the model we derived for undulating surfaces! Other mechanisms must be important; we'll talk about these next lecture.

